

Project 3 Written Portion

Wednesday, April 29, 2026 3:02 PM

1. Gram-Schmidt in $\mathbb{R}^4$ (Four Vectors)

Given the four vectors

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{a}_4 = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \end{bmatrix},$$

apply the Gram-Schmidt algorithm to construct an orthonormal set. Refer to the algorithm provided in the lecture notes.

$$a) \begin{bmatrix} 0.5774 & 0.5774 & 0 & -0.5774 \\ 0.5774 & -0.5774 & 0.5774 & 0 \\ 0 & 0.5774 & 0.5774 & 0.5774 \\ 0.5774 & 0 & -0.5774 & 0.5774 \end{bmatrix}$$

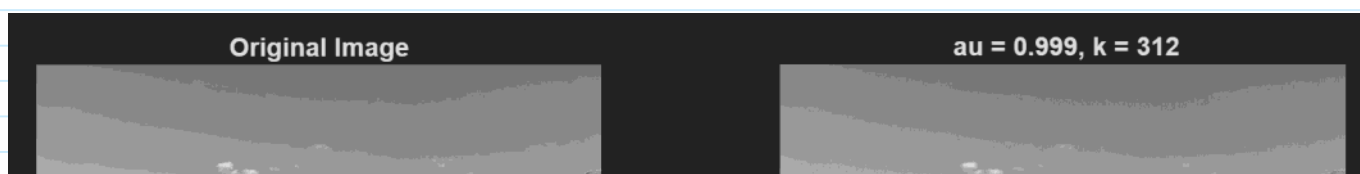
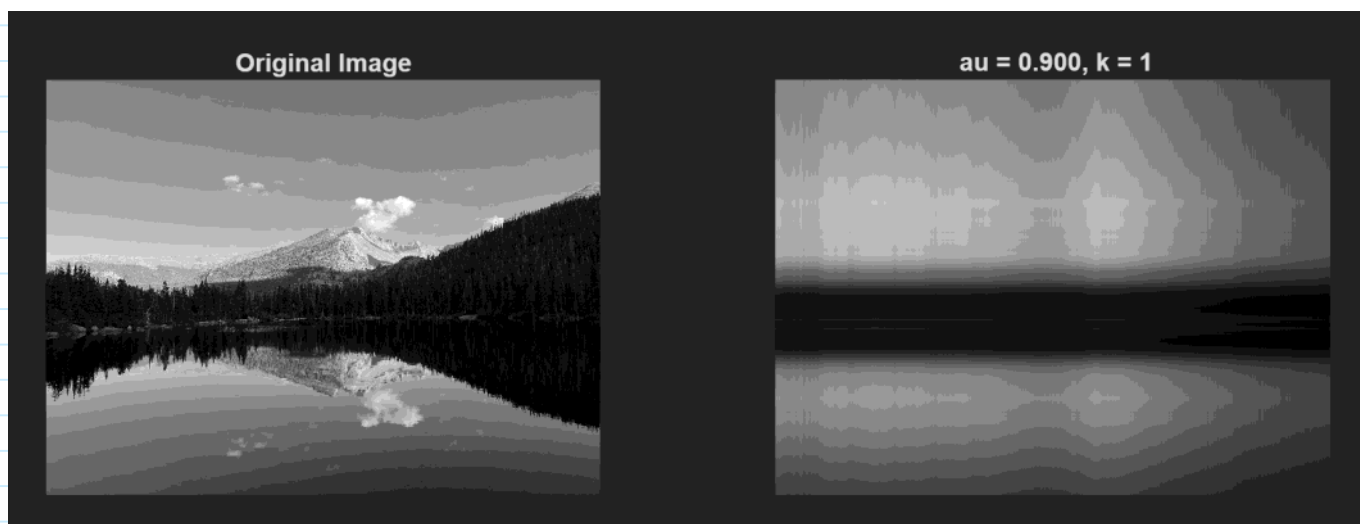
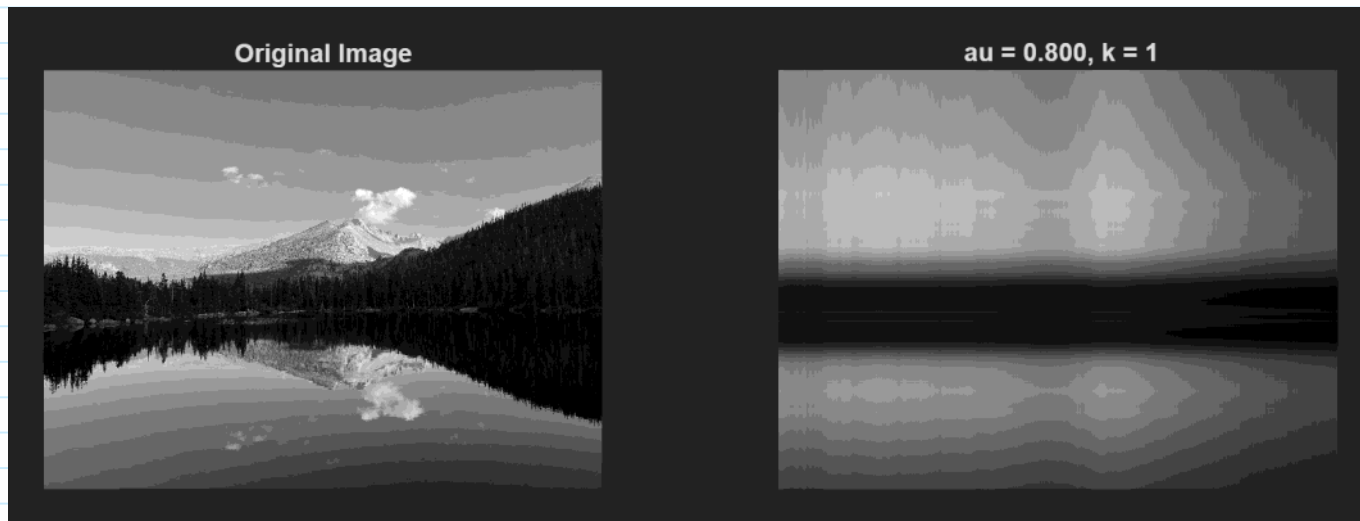
b) The matrix $Q^T Q = \text{identity matrix } (I)$ due to the columns of Q being orthonormal

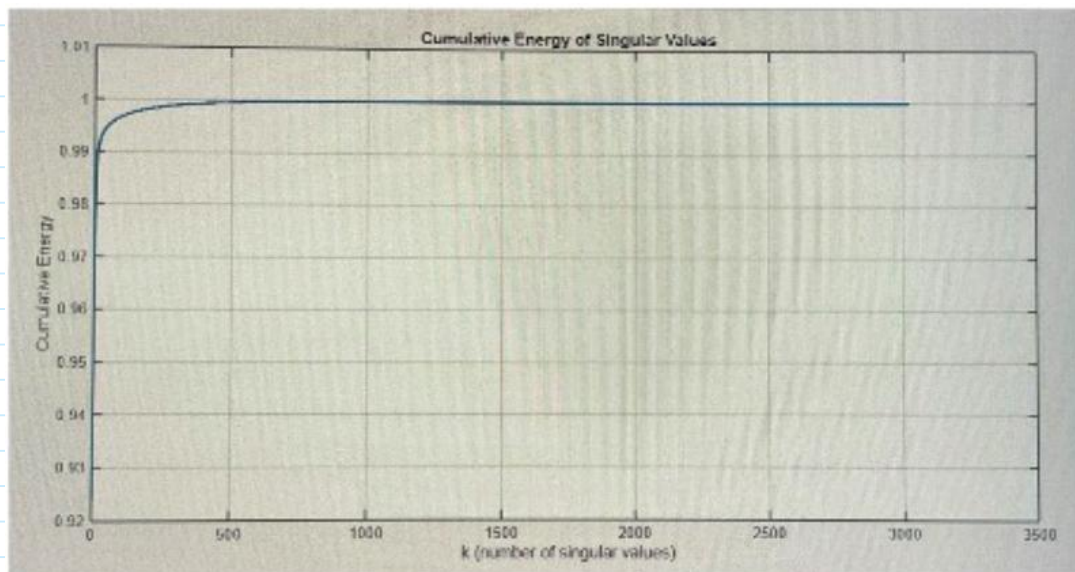
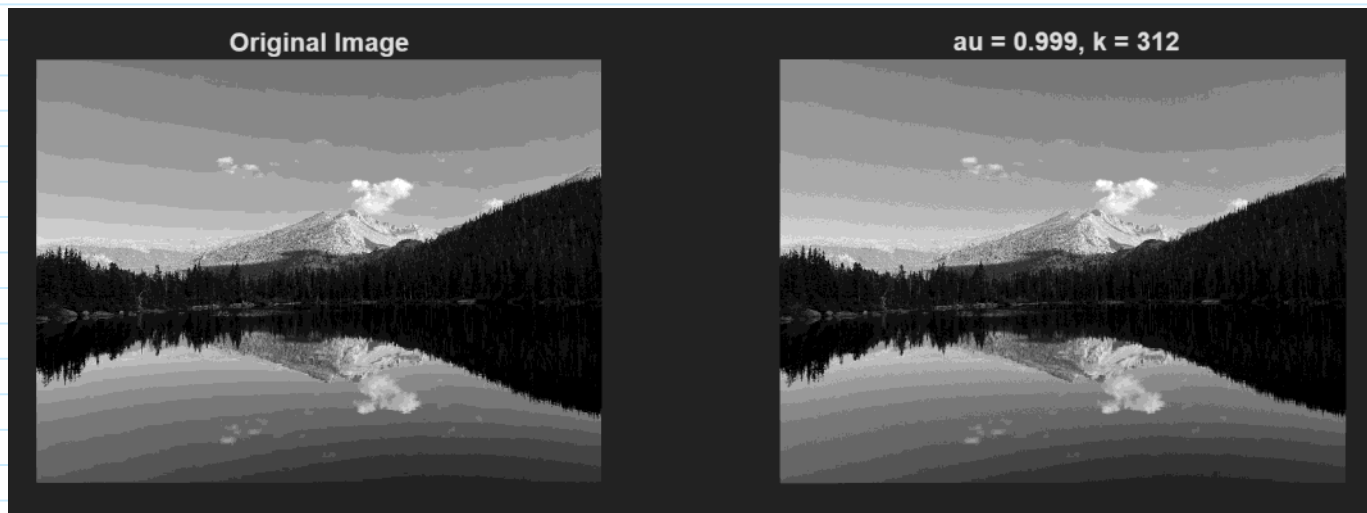
2. Image Compression Using Truncated SVD

In this problem you will write a function to perform truncated SVD and use it to compress an image. The goal is to study how much information ("energy") is captured by the leading singular values.

a) Threshold	Rank
0.80	1
0.90	1
0.99	13
0.999	312

b)





c) As energy threshold increases, rank k also increases.

3. Multiple Linear Regression Using QR Decomposition

You are given electricity usage data for five households. Each entry includes: Temperature ($^{\circ}\text{F}$), Household Size, and Monthly Electricity Usage (kWh).

Temperature ($^{\circ}\text{F}$)	Household Size	Usage (kWh)
55	2	410
60	3	450
70	4	520
75	3	480
80	5	600

We model electricity usage using:

$$\text{Usage} = \beta_0 + \beta_1(\text{Temperature}) + \beta_2(\text{Household Size}).$$

You will use the **QR decomposition** to compute the least-squares solution.

a) The regression system overdetermined due to more equations (5) than unknown parameters (3)

b) $B_0 = 209.3007$

$$B_1 = 1.6364$$

$$B_2 = 50.4196$$

c) Predicted usage = 517.3427 kwh

d) $R^2 = 0.9870 \rightarrow$ indicates regression model fits the data very well